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MOTION NEAR A HOPF BIFURCATION OF A THREE-DIMENSIONAL SYSTEM

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Introduction

In this article, we study the three-dimensional autonomous system treated by Rand [1]. The method of multiple scales [2] is used to obtain an asymptotic expansion for the amplitude and frequency of the limit cycle near the Hopf bifurcation [3] of the system. The resulting analytical approximation is valid only in a small range near the Hopf bifurcation point. This range is ascertained using numerical simulations and stability analysis. Floquet theory [4] is used to determine the stability of the periodic solutions. Also, Rand's analysis is discussed in light of the present study. The perturbation analysis presented here can be easily applied to higher-dimensional systems. The discussion on the range of validity of the analytical approximation and the stability analysis are pertinent to higher-dimensional systems as well.

The equations representing the system are

$$\dot{x} = \mu x - y - xz \tag{1}$$

$$\dot{y} = \mu y + x \tag{2}$$

$$\dot{z} = -z + x^2 z + y^2 \tag{3}$$

where x, y, and z are three scalar variables, μ is a scalar control parameter, and the overdot indicates the derivative with respect to time t. The fixed point of the system $x = y = z = 0$ is stable (unstable) when $\mu < 0 (\mu > 0)$. The Hopf bifurcation of the system occurs at $\mu = 0$. It follows from the form of the equations that if x, y, and z is a solution, then $-x$, $-y$, and z is also a solution. Using a center manifold reduction [3], Rand [1] reduced (1)-(3) into a two-dimensional system. Then, he introduced a near-identity transformation to determine an approximation to the limit cycles near the Hopf-bifurcation point. Here, we use the method of multiple scales to determine a first-order approximation to the limit cycles near the Hopf-bifurcation point and determine its range of validity with the help of numerical simulations and stability analysis. Further, we show that as μ increases from zero, the limit cycle displays an inversion symmetry in the x-y plane, increases in size, and deforms smoothly until μ reaches $\mu^{(1)} \approx 0.300$. As μ increases beyond $\mu^{(1)}$, the

limit cycle loses its inversion symmetry. Then, as μ increases further, the two resulting unsymmetric limit cycles deform smoothly until μ reaches the second critical value $\mu^{(2)}\approx 0.4405$. As μ exceeds $\mu^{(2)}$, the period of the unsymmetric limit cycles doubles. This sequence of bifurcations is expected in symmetric systems [5]. The results show that a general disturbance provided along all directions is necessary for the detection of the instabilities.

Perturbation Analysis

Using the method of multiple scales [2], we seek expansions for the three scalar variables in the form

$$x = \sum_{n=0}^{3} \varepsilon^n x_n(T_0, T_1, T_2) + \ldots \tag{4}$$

$$y = \sum_{n=0}^{3} \varepsilon^n y_n(T_0, T_1, T_2) + \ldots \tag{5}$$

$$z = \sum_{n=0}^{3} \varepsilon^n z_n(T_0, T_1, T_2) + \ldots \tag{6}$$

where $T_n = \varepsilon^n t$ and ε is a small dimensionless parameter that is used as a bookkeeping device and will be set equal to unity in the final analysis. In terms of the T_n, the time derivative becomes

$$\frac{d}{dt} = D_0 + \varepsilon D_1 + \varepsilon^2 D_2 + \ldots \tag{7}$$

where $D_n = \partial/\partial t_n$. Also, the control parameter is expanded as

$$\mu = \varepsilon^2 \mu_2 + 0(\varepsilon^3) \tag{8}$$

where μ is ordered at $0(\varepsilon^2)$ so that the influence of the nonlinear terms and the control parameter μ is realized at the same order in the following analysis.

Substituting (4)-(8) into (1)-(3) and equating coefficients of like powers of ε, we obtain the following equations at different orders of ε:

$0(\varepsilon)$:

$$D_0 x_1 + y_1 = 0 \tag{9}$$

$$D_0 y_1 - x_1 = 0 \tag{10}$$

$$D_0 z_1 + z_1 = 0 \tag{11}$$

$0(\varepsilon^2)$:

$$D_0x_2 + y_2 = -D_1x_1 - x_1z_1 \tag{12}$$

$$D_0y_2 - x_2 = -D_1y_1 \tag{13}$$

$$D_0z_2 + z_2 = -D_1z_1 + y_1^2 \tag{14}$$

$0(\varepsilon^3)$:

$$D_0x_3 + y_3 = \mu_2x_1 - x_1z_2 - x_2z_1 - D_2x_1 - D_1x_2 \tag{15}$$

$$D_0y_3 - x_3 = \mu_2y_1 - D_2y_1 - D_1y_2 \tag{16}$$

$$D_0z_3 + z_3 = x_1^2z_1 + 2y_1y_2 - D_2z_1 - D_1z_2 \tag{17}$$

The nondecaying solution of (9)-(11) is

$$x_1 = iA(T_1, T_2)e^{iT_0} + cc \tag{18}$$

$$y_1 = A(T_1, T_2)e^{iT_0} + cc \tag{19}$$

$$z_1 = 0 \tag{20}$$

where cc stands for the complex conjugate of the preceding terms and A is determined by imposing solvability conditions at the next level of approximation.

Substituting (18)-(20) into (12)-(14) and eliminating the terms that produce secular terms, we have $D_1A = 0$ or $A = A(T_2)$. Then, the solution of (12)-(14) is

$$x_2 = y_2 = 0 \tag{21}$$

$$z_2 = A\bar{A} + \frac{A^2e^{2iT_0}}{1 + 2i} + cc \tag{22}$$

where $\bar{A}$ is the complex conjugate of A.

Substituting (18)-(22) into (15) and (16) and eliminating the terms that produce secular terms, we obtain

$$2A' - 2\mu_2A + \left(\frac{9}{5} + \frac{2}{5}i\right)A^2\bar{A} = 0 \tag{23}$$

where the prime denotes the derivative with respect to T_2. Substituting $A = \frac{1}{2}ae^{i\beta}$, where a and β are real quantities into (23), separating real and imaginary parts, and setting $\varepsilon = 1$, we obtain

$$\dot{a} = \mu_2a - \frac{9}{40}a^3 \tag{24}$$

$$\beta = \frac{-1}{20} a^2 \tag{25}$$

Equations (24) and (25) are equivalent to those obtained by Rand [1] using normal forms. We note that the method of multiple scales involves less algebra than the method of normal forms.

It follows from (24) that $a \to 0$ when $\mu_2 < 0$ and $a \to \frac{2}{3}\sqrt{10\mu_2}$ when $\mu_2 > 0$. For $\mu > 0$, there is an exchange of stability between the limit-cycle branch and the fixed-point branch. Thus, $\mu = 0$ is a supercritical Hopf-bifurcation point [3]. It follows from (4)-(6), (18)-(20), (21), (22), (24), and (25) that, to second order the limit-cycle solution of (1)-(3) is given by

$$x = -a \sin \theta t + 0(a^3) \tag{26}$$

$$y = a \cos \theta t + 0(a^3) \tag{27}$$

$$z = \frac{1}{2} a^2 \left[1 + \frac{1}{5} \cos 2\theta t + \frac{2}{5} \sin 2\theta t \right] + 0(a^3) \tag{28}$$

where

$$a = \frac{2}{3}\sqrt{10\mu} \quad \text{and} \quad \theta = (1 - \frac{1}{20} a^2) + 0(a^3) \tag{29}$$

It should be borne in mind that the expansions (26)-(28) are determined for small values of μ and hence their accuracy is expected to deteriorate as μ becomes large. Next, we compare it with numerical solutions of equations (1)-(3).

Comparison

Figure 1 compares the analytical approximation (dashed lines) with the numerical solutions (solid lines) of equations (1)-(3) for different values of μ. The numerical solutions are obtained by integrating equations (1)-(3) using a Runge-Kutta-Verner fifth- and sixth-order method. From the figure, we can see that the approximation deteriorates as μ increases to 0.2 and beyond. In fact, at $\mu = 0.3$ it is not even close to the numerical solution in either form or magnitude. The analytical approximation is not shown for values of $\mu > 0.3$. Next, we present the stability analysis.

Stability Analysis

To describe the dynamics of the system (1)-(3), we need to determine the stable and unstable limit cycles. These limit cycles are located using an algorithm originally proposed by Aprille and Trick [6]. It starts with a guess for the period and initial conditions of the limit cycle and then uses a combination of a numerical integration scheme and a Newton-Raphson iteration procedure. This algorithm is sensitive to the initial guesses and the step size of the numerical integration when multiple solutions coexist.

Once a periodic solution x(t), y(t), and z(t) with period T has been determined analytically or calculated numerically using the algorithm, we determine its stability by using Floquet theory. To this end, we superimpose on x(t), y(t), and z(t) a general disturbance having the components $\xi_1(t)$, $\xi_2(t)$, and $\xi_3(t)$, respectively, substitute the total solution into (1)-(3), linearize the resultant equations, and obtain

$$\dot{\xi} = \begin{bmatrix} \mu - z & -1 & -x \\ 1 & \mu & 0 \\ 2xz & 2y & x^2 - 1 \end{bmatrix} \xi \tag{30}$$

where ξ is a column vector whose components are ξ_1, ξ_2, and ξ_3 and the 3x3 (Jacobi) matrix on the right-hand side of (30) is periodic with period T. Hence, a solution x(t), y(t), and z(t) of (1)-(3) is asymptotically stable if and only if $\xi(t)$ decays with time. To determine the behavior of $\xi(t)$ as $t \to \infty$, we calculate three linearly independent solutions of (30) from t = 0 to t = T by using the initial conditions

$$\xi_1(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \xi_2(0) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \xi_3(0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{31}$$

Then, we form the monodromy matrix [4]

$$M = [\xi_1(T) \quad \xi_2(T) \quad \xi_3(T)] \tag{32}$$

The eigenvalues of the 3x3 matrix M are called Floquet multipliers and $\xi(t) \to 0$ as $t \to \infty$ if and only if all Floquet multipliers are within the unit circle in the complex plane. For the numerical solutions, the monodromy matrix is obtained as a by-product of the algorithm used to determine the periodic solutions. Moreover, one of the Floquet multipliers is at 1 for all values of μ .

The analytical approximation, equations (26) and (27), corresponds to a circle in the x-y plane. It displays inversion-symmetry in the x-y plane; that is, $x(t) = -x(t + \frac{1}{2}T)$ and $y(t) = -y(t + \frac{1}{2}T)$ where T is the period of the solution. Rand [1] studied the stability of the analytical approximation to a disturbance along the z-direction only; that is, $\xi_1 = \xi_2 = 0$ in equation (30) and found that the solution loses stability at $\mu^{(3)} = 0.45$. However, if one considers a disturbance along all directions, a Floquet multiplier leaves the unit circle at $\mu \approx \mu^{(4)} = 0.194$, implying that the analytical solution is unstable beyond $\mu^{(4)}$.

Considering the stability of the numerical solution to a general disturbance,we find that the solution loses its inversion-symmetry in the x-y plane at $\mu = \mu^{(1)} \approx 0.300$ due to a symmetry-breaking bifurcation. Following this bifurcation, the symmetric solutions of equations (1)-(3) become unstable. Also, the frequency spectra of the x and y variables contain even harmonics. The symmetry-breaking bifurcation is detected by determining the Floquet multipliers of the matrix

$$M_2 = \left[\xi_1(\frac{1}{2}T) \quad \xi_2(\frac{1}{2}T) \quad \xi_3(\frac{1}{2}T) \right] \tag{33}$$

which is obtained by integrating over one-half the period of the limit cycle; that is, $\frac{1}{2}T$. At $\mu = \mu^{(1)}$ a Floquet multiplier corresponding to M_2 leaves the unit circle through 1, indicating the occurrence of the symmetry-breaking bifurcation. For the symmetric solutions, a multiplier always stays at -1. If, we had considered the stability of the numerical solution to a disturbance along the z-direction only, then, we would not have detected this bifurcation.

As μ increases beyond $\mu^{(1)}$, the two clone unsymmetric limit cycles deform smoothly (one of these clones is shown in Figures 1d. 1e, and 1f; the other can be obtained by replacing x and y with -x and -y, respectively) until μ reaches the second critical value $\mu^{(2)} \approx 0.4405$. At this value, a Floquet multiplier corresponding to matrix M leaves the unit circle through -1 indicating the occurrence of the first period-doubling bifurcation.

Rand [1] used the analytical approximation, discussed earlier, to predict the first period-doubling bifurcation. The form of the analytical approximation is incorrect for the unsymmetric solutions following the symmetry-breaking bifurcation. Moreover, in the present case, this bifurcation occurs at a value

of μ which is beyond the range of validity of the analytical approximation. So, the analytical approximation is not good enough to predict the symmetry-breaking bifurcation and the subsequent bifurcations.

Concluding Remarks

We have illustrated the use of the method of multiple scales to obtain asymptotic expansions for motions near a Hopf bifurcation of a three-dimensional system. This perturbation analysis can be easily extended to higher-dimensional systems. The motion of the reduced system following the center manifold reduction usually displays an inversion-symmetry. In those cases, it is expected that the first period-doubling instability will be preceded by a loss of symmetry through a symmetry-breaking bifurcation. If the analytical approximation is symmetric, it can at most be used to detect the symmetry-breaking bifurcation preceding the first period-doubling bifurcation. In the stability analysis, a disturbance along all directions is necessary to detect the symmetry-breaking bifurcation.

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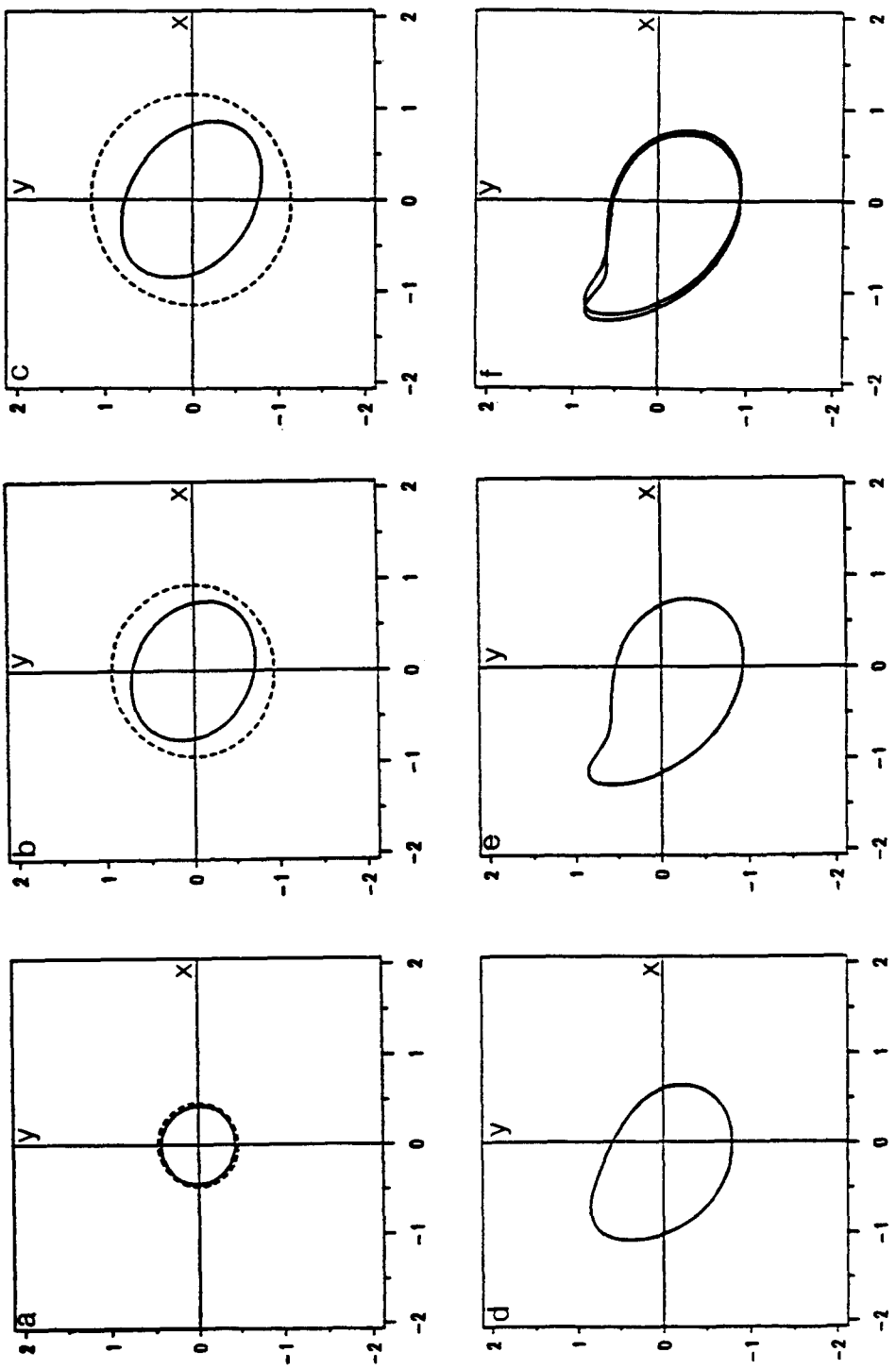


Figure 1. Solutions of equations (1) to (3): (a) $\mu = 0.05$, (b) $\mu = 0.2$, (c) $\mu = 0.3$, (d) $\mu = 0.35$, (e) $\mu = 0.44$, and (f) $\mu = 0.445$; ____ numerical integration, - - - - analytical approximation.